Econ 470 HW 5.3

Claire Hamilton

Summarize the Data

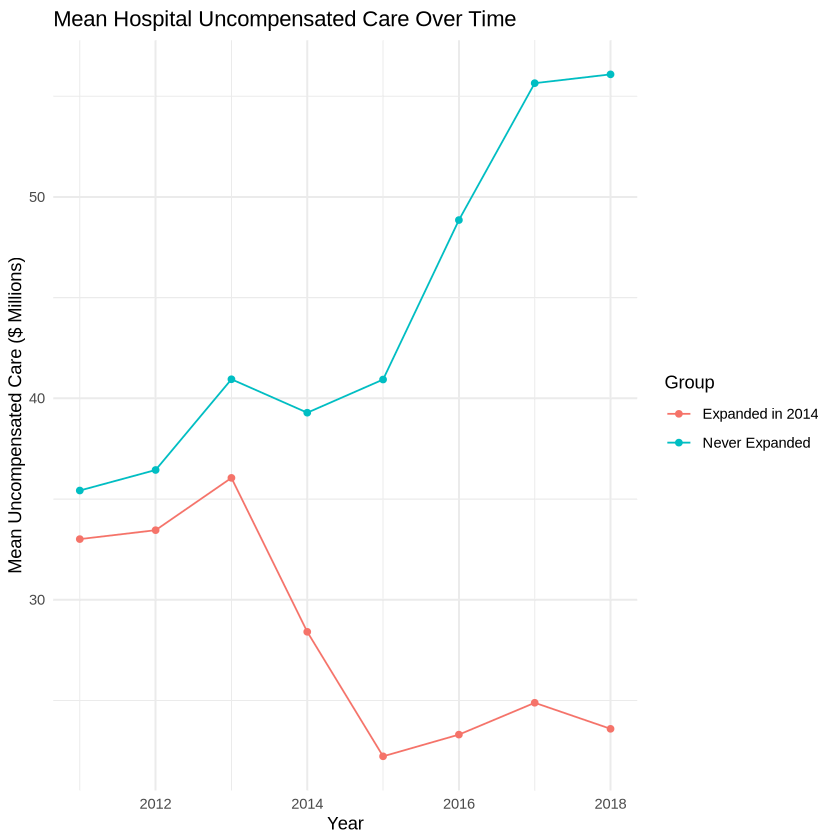
Question 1

A tibble: 8 × 2

Year	Mean Uncompensated Care ($ Millions)
<dbl>	<dbl>
2011	34.32
2012	37.02
2013	39.15
2014	36.37
2015	32.94
2016	44.52
2017	40.49
2018	38.03

Mean hospital uncompensated care gradually increased from 2011-2013, then saw a drop in 2014-2015. There is also a slight spike in uncompensated care in 2016-2017.

Question 2



The graph does suggest a potential treatment effect, as uncompensated care declined starting in 2014 for the expansion states, while it increased substantially for non-expansion states. This suggests that Medicaid expansion could be associated with a decrease in hospital uncompensated care.

Question 3

A tibble: 2 × 3

Group	2012 (Pre)	2015 (Post)
<chr>	<dbl>	<dbl>
Expansion (2014)	35.03	22.23
Non-Expansion	43.49	45.62

Question 4

My results show that states that expanded Medicaid in 2014 saw substantial declines in uncompensated care, while states that never expanded saw increases over the same time period. This pattern is consistent with the Affordable Care Act's Medicaid expansion policy, which extended Medicaid eligibility to low-income adults (up to 138% of the Federal Poverty Level) and increased enrollment through federal funding. Since coverage was expanded to previously uninsured patients, hospitals in expansion states likely treated more patients with insurance rather than providing care

without payment. Thus, uncompensated care decreased because hospitals were more likely to be reimbursed through Medicaid rather than bearing the costs of treating uninsured patients.

Estimate ATEs

Question 5

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Table: DD Effect of Medicaid Expansion on Uncompensated Care

|                  |     (1)    |
|:-----------------|-----------:|
| Post             |  14.195*** |
|                  |   (1.475)  |
| Expansion State  |   -3.474*  |
|                  |   (1.511)  |
| Post × Expansion | -22.629*** |
|                  |   (1.858)  |
| Num.Obs.         |    16903   |
| R2               |    0.034   |
| F                |   196.072  |

__Note:__
^^ + p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001
```

The DD estimate of the effect of Medicaid expansion on hospital uncompensated care is around -22.6 million dollars. This indicates that, after 2014, hospitals in expansion states had a reduction in uncompensated care of about 22.6 million dollars relative to the change seen in non-expansion states.

Question 6

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Table: Effect of Medicaid Expansion on Uncompensated Care (TWFE)

|                         |     (1)    |
|:------------------------|-----------:|
| Medicaid Expansion TWFE | -26.853*** |
|                         |   (4.129)  |
| Num.Obs.                |    16903   |
| R2                      |    0.838   |
| Std.Errors              |  by: state |
| FE: provider_number     |      X     |
| FE: fyear               |      X     |

__Note:__
^^ + p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001
```

When including hospital and year fixed effects, the estimated effect of Medicaid expansion on uncompensated care is about −26.8 million dollars. This is similar to the estimate from Question 5 (−$22.6 million) but larger in magnitude, suggesting an even stronger effect after controlling for differences between hospitals and changes over time. Overall, both models suggest that Medicaid expansion substantially reduced hospital uncompensated care.

Question 7

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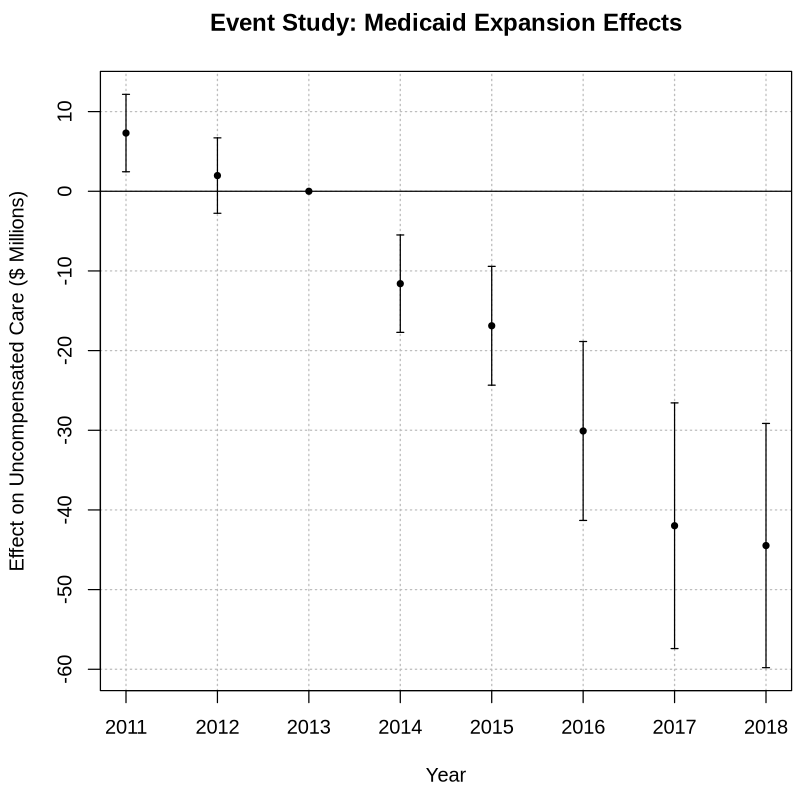
Table: Effect of Medicaid Expansion on Uncompensated Care (All States)

|                           |     (1)    |
|:--------------------------|----------:|
| Medicaid Expansion Effect | -22.508*** |
|                           |   (4.072)  |
| Num.Obs.                  |    22852   |
| R2                        |    0.846   |
| Std.Errors                |  by: state |
| FE: provider_number       |      X     |
| FE: fyear                 |      X     |

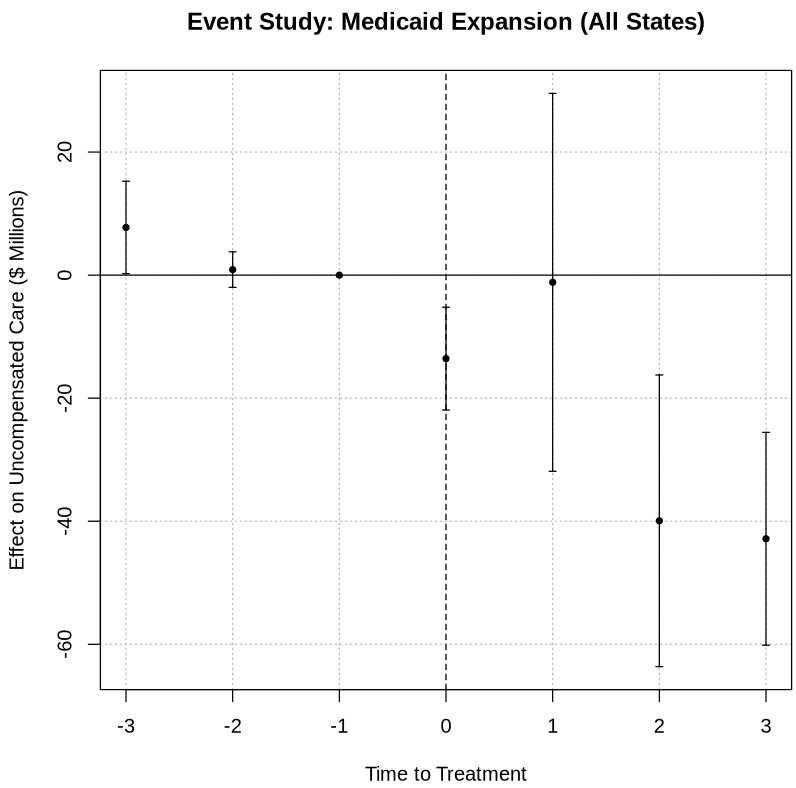
__Note:__
^^ + p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001
```

When including all states, the estimated effect of Medicaid expansion is about −22.5 million dollars, which is smaller in magnitude but similar to the estimate from Question 6 (−26.8 million dollars). The results are different because the control group now includes states that expanded later, not just states that never expanded. These states are likely more similar to expansion states, which reduces the difference between the treated and control groups and leads to a smaller estimated effect.

Question 8


Event Study: Medicaid Expansion Effects
Effect on Uncompensated Care ($ Millions)
10
0
-10
-20
-30
-40
-50
-60
2011
2012
2013
2014
2015
2016
2017
2018
Year

Question 9


Event Study: Medicaid Expansion (All States)
Effect on Uncompensated Care ($ Millions)
20
0
-20
-40
-60
-3
-2
-1
0
1
2
3
Time to Treatment

Question 10

Medicaid expansion is associated with a large and statistically significant reduction in hospital uncompensated care, around $22–27 million per hospital across the different models. The event study graphs show clear declines after expansion, though the pre-treatment confidence intervals do not all include zero, particularly in 2011 for the common treatment timing event study. This initially raised minor concerns about parallel trends, however, the event study with staggered treatment timing has pre-treatment effects that are closer to zero. Overall, the evidence strongly suggests that Medicaid expansion reduced hospital uncompensated care.

A limitation of the standard TWFE approach is that it uses states that expand later as controls and combines multiple comparisons across different treatment timings. Because these groups may differ and treatment effects can change over time, this can lead to bias in the estimated effect and make it harder to interpret.